

The Citizens Protein Project 2

The first publicly crowd-funded observational study on exhaustive analysis of popular whey protein supplements in India reveal poor quality and deceptive marketing claims of medical pharmaceutical- compared to nutraceutical- industry powders

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Abstract

This uniquely original, crowd-funded laboratory study compared 18 medical-pharmaceutical and 16 nutraceutical whey protein powders sold in India by analyzing macronutrients, amino-acid profiles, sugars, heavy metals, mycotoxins, hormones, pesticides and undeclared additives. Nutraceutical products supplied a mean ($\pm$ SD) 75.6 ± 7.3 g protein/100g and showed tight agreement with label claims. Ninety-four per cent met the ≥ 60 g/100g threshold and exhibited high essential-amino-acid density (up to 8.5g leucine/100g) with negligible added sugars. By contrast, pharmaceutical powders averaged 29.1 ± 15.4 g protein/100g; 83% were mis-labeled beyond $\pm 5\%$, and only one product reached the quality benchmark. None exceeded 5g leucine/100g, while 44% contained ≥ 2 g sucrose or fructose/100g, often unlabeled. Heavy-metal outliers (cadmium, arsenic, copper), aflatoxin, and progesterone appeared more frequently in pharmaceutical brands, and taurine-based nitrogen spiking was detected in 89% of them versus 12.5% of nutraceuticals. No product contained detectable pesticides or mercury. Overall, nutraceutical powders consistently met compositional expectations, whereas medical-pharmaceutical formulations delivered lower-quality protein, higher sugar loads and unreliable labeling, questioning their suitability for therapeutic use.

Abbreviations: EAA = essential amino acids.

Keywords: dietary supplements, fitness, heavy metals, lifestyle, mislabeling, nutrition

1. Introduction

Recent peer-reviewed investigations of whey and other protein powders revealed substantial quality and safety deficits. A 2024 investigation from India analyzing 36 widely available products found that 75% contained lead, 27.8% cadmium, and 13.9% aflatoxins. Several products harbored pesticide residues and volatile organic compounds.^[1] Most products failed to match their labeled protein content, some falling short and others exceeding the claims raising credible suspicion of “nitrogen or

amino-acid spiking,” a deceptive practice used to inflate protein readings.^[1,2]

Quantitative risk assessments using large datasets confirm that heavy metals are commonly detectable in protein powders. Yet, formal exposure limits for these elements in supplements are inconsistently defined, and rarely enforced.^[3,4] This is particularly troubling given the toxicological context: the World Health Organization and other public health bodies emphasize that there is no safe level of lead exposure, underscoring why even “trace” concentrations merit disclosure.^[5,6]

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The authors have no conflicts of interest to disclose.

The datasets generated during and/or analyzed during the current study are publicly available.

Because this study did not involve human participants, the Rajagiri Hospital's institutional review board determined that its approval was not required.

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Despite these risks, regulatory transparency remains inadequate. In the United States, the Food and Drugs Administration, does not approve dietary supplements before they enter market, typically intervening only after products are on shelves.^[7] In India, Food Safety and Standards Authority of India prohibits hormones or steroids in nutraceuticals and mandates detailed labeling of use and target population. However, product quality data are submitted by manufacturers and are not made publicly available, perpetuating opacity.^[1,7,8]

Recognizing these gaps, Indian government panels have recently proposed shifting some nutraceutical oversight from Food Safety and Standards Authority of India to the drug regulator (CDSCO), to curb exaggerated claims.^[9,10] Collectively, these findings indicate that whey protein supplements—sold to athletes, patients, and the public, operate in a marketplace marked by contamination, mislabeling, and poor disclosure. Regulatory frameworks provide limited pre-market verification and scant public access to analytical data. Therefore, establishing transparent, publicly accessible databases of independent test results is both a consumer-protection necessity and an ethical imperative for evidence-based nutrition.

To address these systemic shortcomings, our study adopted a novel “citizen science” framework. By securing funding directly from the public, we established a research model free from industry influence and institutional biases, ensuring complete transparency and objectivity. This crowd-funded approach is not merely a funding mechanism but a core component of our methodology, designed to generate an unbiased, publicly accessible database of independent test results. This positions our work as a direct response to the prevailing lack of transparency, making it a crucial tool for consumer protection, evidence-based nutrition, and regulatory accountability.

In this context, we systematically assayed 34 whey protein powders available in Indian market, comparing “medical pharmaceutical” products prescribed to patients, with “nutraceutical” whey supplements. We aimed to quantify true protein yield and comprehensively assess multiple quality parameters, including contamination with heavy metals, aflatoxins, and hormones. From a public-health perspective, the study also evaluated economic and clinical burden of medical pharmaceutical-grade powders, identified deceptive labeling practices, and generated crowd-funded, openly accessible data to protect patients and inform regulators and consumers.

2. Methods

2.1. Product purchase and preparation

We exhaustively reviewed online supplements stores and authorized protein supplements wholesale and retail sellers to prepare a list of popular medical pharmaceutical and nutraceutical industry marketed highest selling whey supplements. Once the list was formulated, the products were purchased directly from manufacturer stores (online or offline) and from authorized dealers. Where applicable, the authenticity of products was confirmed using QR code instructions provided. A designated person was independently directed to prepare a master chart (Supplementary Document 1, Supplemental Digital Content, <https://links.lww.com/MD/Q662>) of purchased products which included name of product, product image, listing and image of disclosed ingredients, nutritional information, and batch number, manufacturing date and expiry information. Upon purchase, all products were maintained in their original, sealed packaging and stored in a climate-controlled, dark, and dry environment to prevent degradation. Expert personnel from an independent, government and multi-regulatory authorized analytical lab (Eureka Analytical Services, India) sampled each coded products under sterile precautions at source, using industry defined protocols and maintaining a reasonable chain of command.

The samples were then transported directly to the designated analytical laboratory, maintaining a strict chain of custody to ensure sample integrity prior to analysis. Once the products were received at the main laboratory, they were processed per standard operating protocols. Because this study did not involve human participants, the Rajagiri Hospital's institutional review board determined that its approval was not required.

2.2. Protocols for product analyses

All products were subjected to analysis as per standards laid down for whey protein supplements testing, by the Bureau of Indian Standards and Association of Official Analytical Chemists. These include total protein content (using Kjeldahl method), detection and quantification of total aflatoxins and aflatoxins G2, B2, B1, and G1 (high pressure liquid chromatography and fluorescence detection), pesticide residue detection and estimation (gas and liquid chromatography and tandem mass spectrometry), detection and quantification of heavy metals (arsenic, cadmium, lead, mercury, and copper by inductively coupled plasma mass spectrometry), sucrose and fructose content and complete amino acid profiling (liquid chromatography with mass spectrometry), taurine quantification (liquid chromatography with fluorescence detection), steroid and hormones detection, and complete profiling of organic and inorganic contents (gas chromatography coupled to mass spectrometry).

3. Results

Of the 34 whey protein supplements analyzed, 18 were marketed as medical–pharmaceutical products and 16 as nutraceuticals.

3.1. Total protein content and mislabeling:

Protein density differed markedly between categories. Nutraceutical products delivered a mean of 75.6 g protein per 100 g powder (median 77.2 g; range 57.2–86.8 g), with 15 of 17 meeting the ≥ 60 g/100 g “high-protein” criterion and only one falling marginally below (57.2 g/100 g). In contrast, pharmaceutical products averaged 29.1 g per 100 g (median 21.5 g; range 10.1–63.8 g), with only one product qualifying as high-protein and the remaining 17 qualified only to be classified as low-protein formulations. Nutraceutical products generally approached or exceeded “Whey Protein Concentrate-80/Whey Protein Isolate” specifications, whereas most pharmaceutical products were substantially diluted with non-protein solids, limiting their clinical utility for delivering clinically meaningful protein doses.

Comparison of manufacturer-declared versus laboratory-measured protein content revealed a stark dichotomy between the 2 marketed groups. In the nutraceutical group, mean bias was -2.7% (slight understatement) with a mean absolute error of 4.7% (median 3.7%). Six products (37.5%) exceeded the $\pm 5\%$ tolerance limit, all due to modest short fills, with the largest deficit being -17% and no overstatements beyond regulatory limits. By contrast, pharmaceutical products displayed far greater variability, with a mean bias of $+2.0\%$ and a mean absolute error of 16.0% (median 9.6%); 15 of 18 (83.3%) breached the $\pm 5\%$ tolerance. Overstatements were common (13/18), reaching up to $+20.8\%$, while 2 products substantially under-delivered, including one providing only 25% of its declared protein content (-74.7%). Overall, nutraceutical products were generally conservative and compliant, while pharmaceutical products demonstrated widespread, clinically relevant mislabelling (Fig. 1). In practical terms, these results indicate that while nutraceutical products were reliably labeled, the labeling of pharmaceutical products was highly erratic, creating a significant risk of patients being either under-dosed or misled about the protein content.

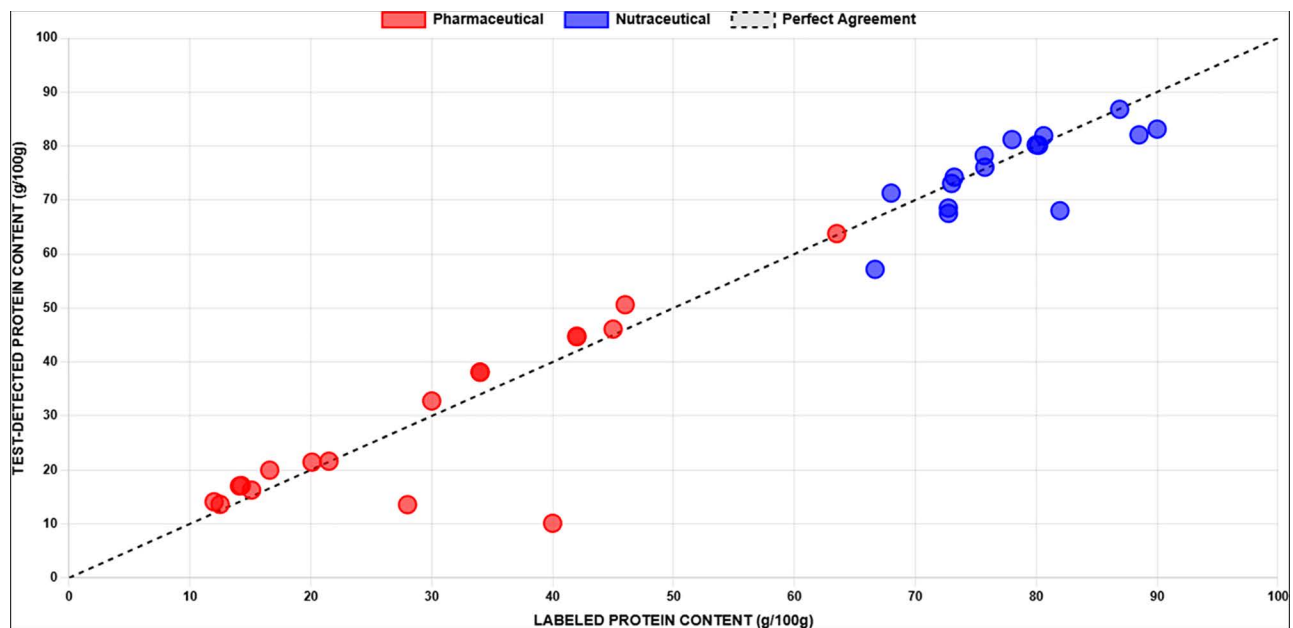


Figure 1. Comparison of labeled versus test-detected protein content in pharmaceutical and nutraceutical protein supplements. Each point represents an individual product, with the diagonal line indicating perfect agreement between labeled and tested values. Red points indicate pharmaceutical products with tested protein content below labeled claims. Blue points indicate nutraceutical products showing higher absolute protein content but greater discrepancies between labeled and tested values.

3.2. Amino acid and sugar and profiles:

Protein quality of whey products was categorized based on the following:

- A. Total essential amino acids (EAA) per 100 g powder–
 - Very high: ≥ 40 g EAA/100 g and leucine ≥ 7 g/100 g
 - High: 30–39.9 g EAA/100 g or leucine 5–6.99 g/100 g
 - Low: 20–29.9 g EAA/100 g or leucine 3–4.99 g/100 g
 - Very low: < 20 g EAA/100 g or leucine < 3 g/100 g
- B. Added sugar (sucrose or fructose)–

Sucrose or fructose ≥ 2 g/100 g dropped one category; ≥ 5 g/100 g dropped 2 while < 0.5 g/100 g was considered essentially no added sugar and high quality.

Nutraceutical products had a mean total amino acid content of 82.26 g/100 g compared to just 25.84 g/100 g in pharmaceutical products (>3-fold difference). EAA content averaged 36.83 g/100 g in nutraceutical products versus 10.98 g/100 g in pharmaceutical (3.4-fold difference). Leucine content ranged from 5.7 g to 8.5 g per 100 g protein. Six powders reached or exceeded both prespecified targets of ≥ 40 g summed EAAs and ≥ 7 g leucine, qualifying as very-high-quality whey isolates.

Pharmaceutical products had consistently lower EAA levels; most contained ≤ 4.9 g leucine/100 g protein, and over half delivered < 20 g EAAs, placing them in the “very low” quality category. Many contained simple sugars in substantial amounts. For example, we detected only 0.61 g leucine in B-protein while it contained 4.83 g sucrose. Similarly, Ensure High-Protein contained only 2.78 g leucine where as Protinex had 3.43 g fructose. No product in this group met the ≥ 7 g leucine or ≥ 30 g total EAA benchmarks, with over half falling below 20 g total EAAs and were therefore classified as very-low quality. Several showed elevated glycine + alanine (> 6 g), indicating inclusion of collagen or non-whey proteins. Thus, nutraceutical isolates generally met accepted criteria for high-quality protein, whereas pharmaceutical powders often provided caloric load without adequate EAA profiles for muscle protein synthesis (Fig. 2).

Sucrose was detected in 44% of pharmaceutical products. Five (27.8%) pharmaceutical products did not label sucrose but had detectable sucrose in the powders. Similarly, 33% of medical pharmaceutical products contained fructose with 27.8%

undeclared. In the nutraceutical group, added sugars were typically below the analytical limit of 0.5 g/100 g protein; only one (Jan Aushadi Whey) exceeded 2 g, raising concerns regarding labeling and quality control.

3.3. Heavy metal content

Heavy metal contamination patterns also differed. Pharmaceutical products showed a higher contamination rate versus nutraceuticals (38% vs 25%, $P < .05$). Mercury was absent in all samples; cadmium and arsenic were the most common contaminants (Fig. 3). Copper was detected in 75% (12 out of 16) of nutraceuticals (range 0.531–4.4 mg/kg; mean 1.68 mg/kg) and in 78% (14 out of 18) of pharmaceuticals, with a wider range (0.529–33.056 mg/kg). Three pharmaceutical products contained ≥ 10 mg/kg copper, while 4 were below the limit of quantification.

3.4. Pesticide residue and fungal toxins, steroids, and hormones

None of the 16 nutraceutical or 18 pharmaceutical powders tested positive for pesticide and insecticide residue. However, fungal toxin (aflatoxin) was detected in 2 pharmaceutical formulations (B-Protin and D-Protin, 11%). Similarly, 2 pharmaceutical powders (B-Protin and Lamino HP, 11%) contained detectable levels of progesterone. Progesterone was also detected in one nutraceutical whey protein. (Naturaltein, 6%)

3.5. Undisclosed taurine addition

Taurine, a non-essential amino acid sometimes used to inflate nitrogen-based protein estimates, was detected in 89% of pharmaceutical and 12.5% of nutraceutical products. Pharmaceutical products contained highly variable amounts (2.30–100.7 mg/100 g; mean 25.8 ± 23.8 mg), with Fresubin High Protein as an outlier at the upper extreme. Per 30-g serving, taurine contributions ranged from 0.7 to 30 mg. Intra-company variability was also pronounced, for example, 4 Abbott India formulations spanned 9.3 to 32.8 mg per

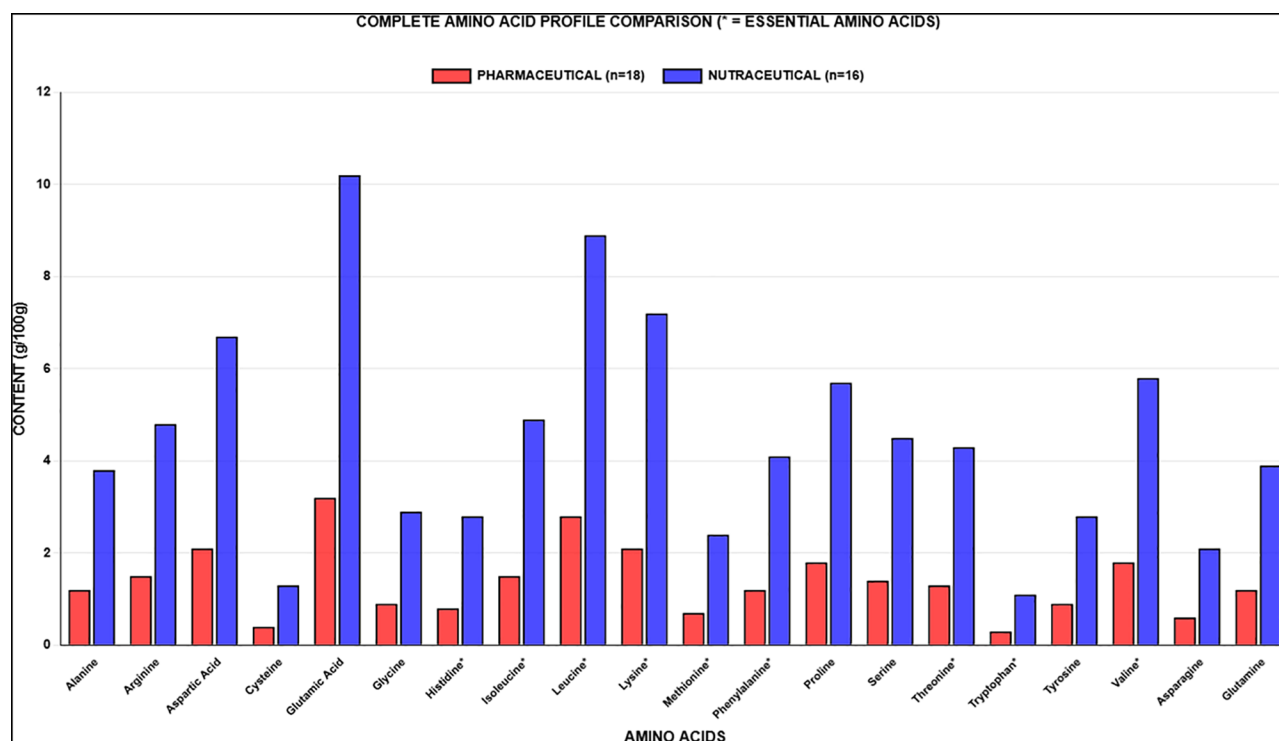


Figure 2. Comprehensive amino acid profile analysis comparing pharmaceutical (n = 18) and nutraceutical (n = 16) protein supplements across all 20 standard amino acids. Essential amino acids are marked with asterisks (*). Values represent mean content (g/100g). Nutraceutical supplements provide vastly superior amino acid profiles compared to pharmaceutical products. The data reveals that nutraceutical products deliver, on average, over 3 times more total EAAs than their pharmaceutical counterparts. Notably, the content of branched-chain amino acids (BCAAs: leucine, isoleucine, valine), which are critical for stimulating muscle protein synthesis, is also over 3-fold higher in nutraceuticals. This profound difference in both the quantity and quality of amino acids underscores the superior nutritional value of nutraceutical formulations for supporting muscle health and recovery.

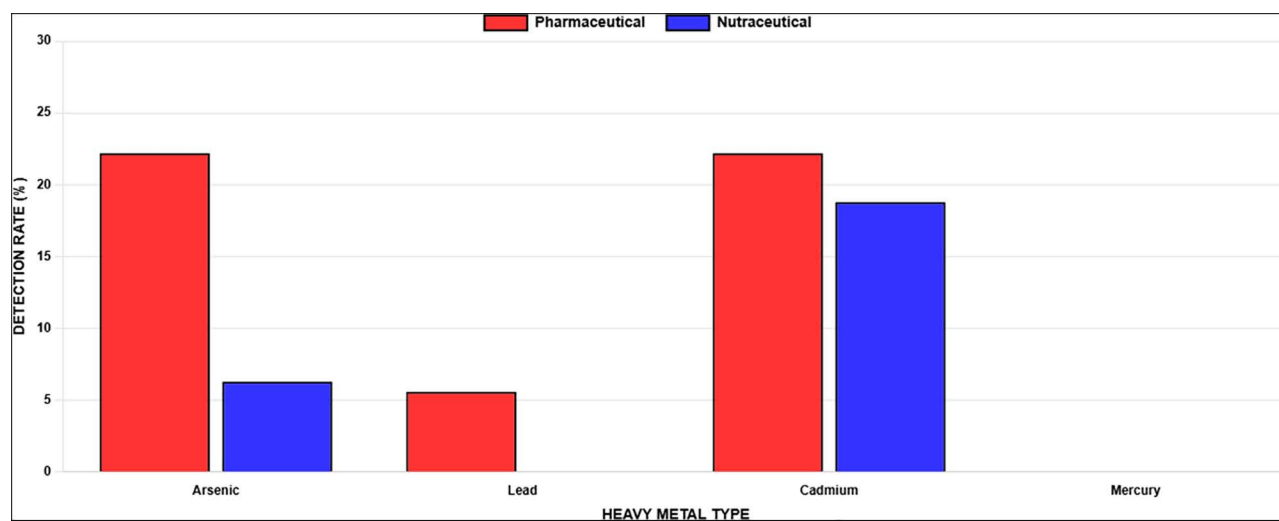


Figure 3. Detection rates of heavy metals in pharmaceutical and nutraceutical protein supplements. Values represent percentage of products with detectable levels above the limit of quantification (LOQ). Pharmaceutical products demonstrated significantly higher overall contamination rates. Arsenic contamination was 3.5-fold higher in pharmaceutical products. Lead was detected only in pharmaceutical products. No mercury was detected in any product tested.

100g, underscoring that taurine inclusion is formulation-specific rather than a corporate standard. Five products fell below 10 mg per 100g, 6 clustered in a moderate band of 10 to 30 mg per 100g, and 5 exceeded 30 mg per 100g, yet even the highest category remained 2 orders of magnitude beneath doses shown to modulate clinical end-points. Consequently, these powders are unlikely to serve meaningful taurine supplementation in healthy adults; their impact would matter only in patient

populations whose protein powder constitutes a major or sole nutrient source. Of the 2 nutraceutical products with added taurine, one contained 5.9mg per 100g while the other had 87.1mg per 100g which were undisclosed on the respective product labels (Fig. 4). The key implication of these findings is not that taurine was added in clinically significant doses, but that its widespread and undisclosed presence in pharmaceutical products points to the use of low-cost, non-protein nitrogen

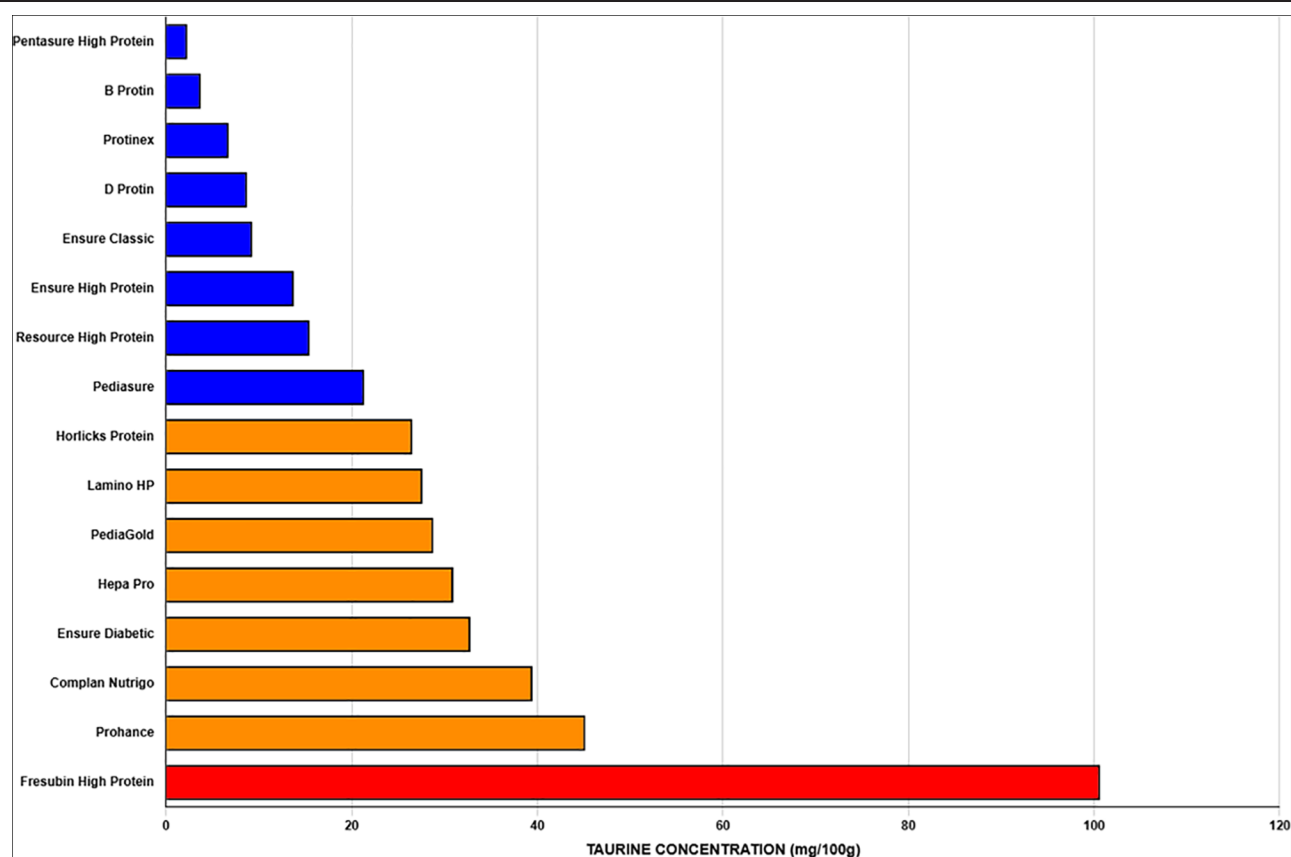


Figure 4. Unlabelled and high taurine content in tested medical pharmaceutical products showing evidence for lower quality of protein.

sources. This practice, often referred to as “nitrogen spiking,” raises ethical concerns regarding manufacturing transparency and the potential to mislead consumers about the true protein quality.

3.6. Cost-effectiveness:

The protein content characteristics between the 2 categories presented a striking contrast. Pharmaceutical products demonstrate significantly lower protein concentrations, with tested values averaging 29.10g per 100g of product. This category exhibits a relatively narrow range of protein content, spanning from as low as 10.1 to a maximum of 63.8g per 100g. In comparison, nutraceutical products show substantially higher protein concentrations, averaging 75.6 grams per 100g in tested values, with a wider range extending from 57.2 to 86.8g per 100g. This fundamental difference in protein density has cascading effects on cost-effectiveness and consumer value propositions.

The accuracy of label claims presented another critical dimension of comparison. While pharmaceutical products showed an average discrepancy of -0.38g per 100g between labeled and tested values, suggesting minimal but consistent under-delivery, nutraceutical products demonstrated larger absolute discrepancies with an average difference of -2.15g per 100g. Despite the larger absolute discrepancies in nutraceutical products, their higher baseline protein content meant they still delivered more protein per serving than most pharmaceutical alternatives. The analysis revealed that 55.6% of pharmaceutical products failed to meet their labeled protein content, while this figure rose to 68.8% for nutraceutical products, indicating systemic quality control issues across both categories.

Cost-effectiveness patterns diverged dramatically between the 2 categories (Fig. 5). Pharmaceutical products generally commanded higher prices per gram of actual protein, ranging from

₹6.71 to ₹24.77 per gram based on tested values. This pricing structure became even more pronounced when actual protein content was considered, with several products showing cost increases exceeding 100% when calculated using tested versus labeled protein values (Fig. 6). The premium pricing of pharmaceutical products appears disconnected from their actual protein delivery, suggesting that other factors beyond protein content influence their market positioning. Conversely, nutraceutical products demonstrate superior cost-effectiveness overall, with prices ranging from ₹2.19 to ₹8.79 per gram of actual protein. This more competitive pricing structure better aligns with protein content and reflects a market focused primarily on protein delivery as the core value proposition.

4. Discussion

In this comparative analysis of 34 whey-protein powders, the first-of-its kind publicly crowd-funded transparent report on popular protein brands categorized into medical pharmaceutical products and nutraceutical industry products sold in the Indian market revealed that products marketed as nutraceutical supplements were superior in multiple aspects (Fig. 7). Nutraceutical protein powders contained a mean protein concentration of $\sim 76\text{g}$ per 100g, showed minimal discrepancies between declared and measured values, offered a high EAA density, contained negligible added sugars, and exhibited fewer contaminant issues. By contrast, powders promoted as pharmaceutical or medical-nutrition products averaged $\sim 29\text{g}$ protein per 100g, were mislabeled in 83 % of cases, displayed inferior EAA profiles (leucine $\leq 4.9\text{g}$ per 100g), contained higher levels of simple sugars, had more frequent heavy-metal exceedances, and occasionally harbored aflatoxin or progesterone. Widespread addition of low-cost taurine in this group further inflated nitrogen-based protein estimates. Collectively, the pharmaceutical

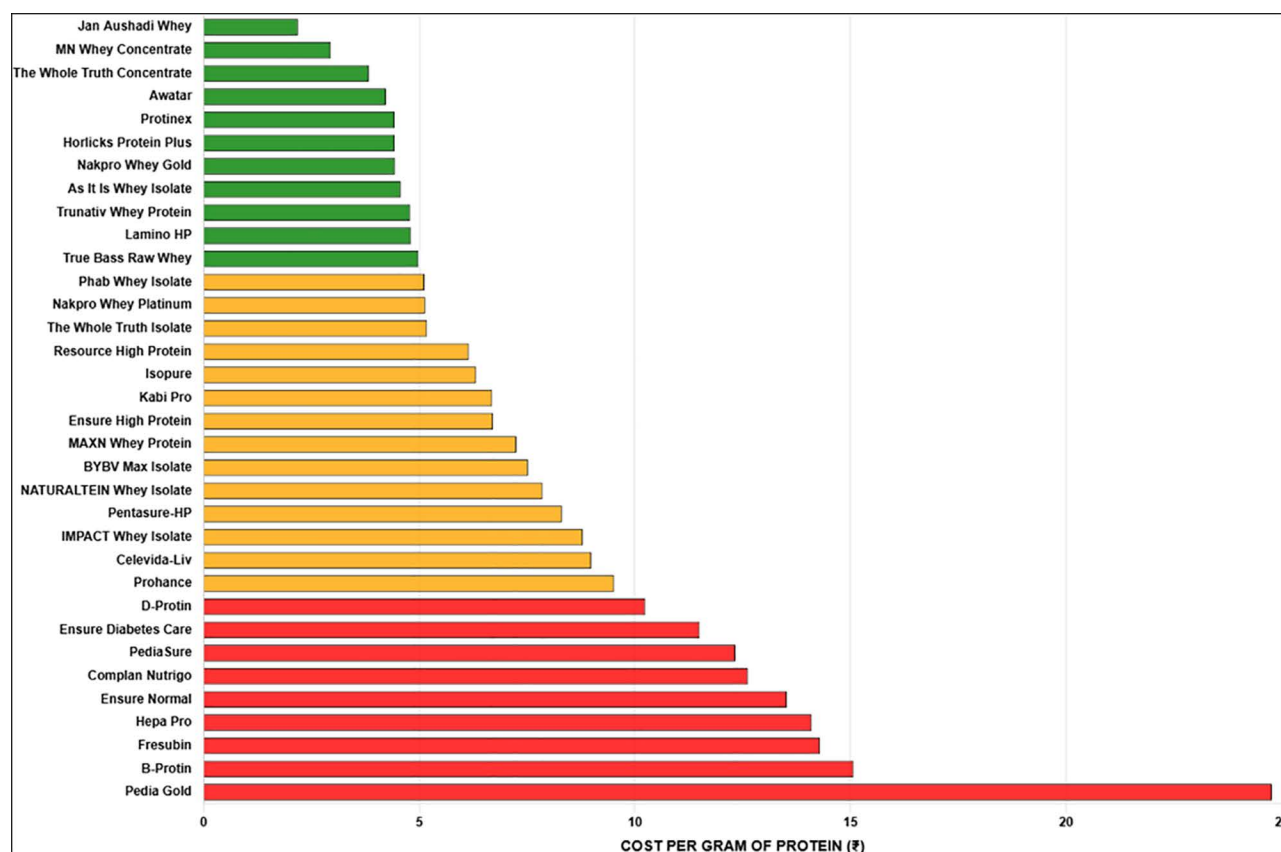


Figure 5. Cost per gram of protein (Indian Rupees) calculated using test-detected protein content for pharmaceutical and nutraceutical products. Products are arranged from most to least cost-effective. Green bars indicate products with cost $\leq$ ₹5/g protein, yellow bars ₹5–10/g, and red bars $>$ ₹10/g.

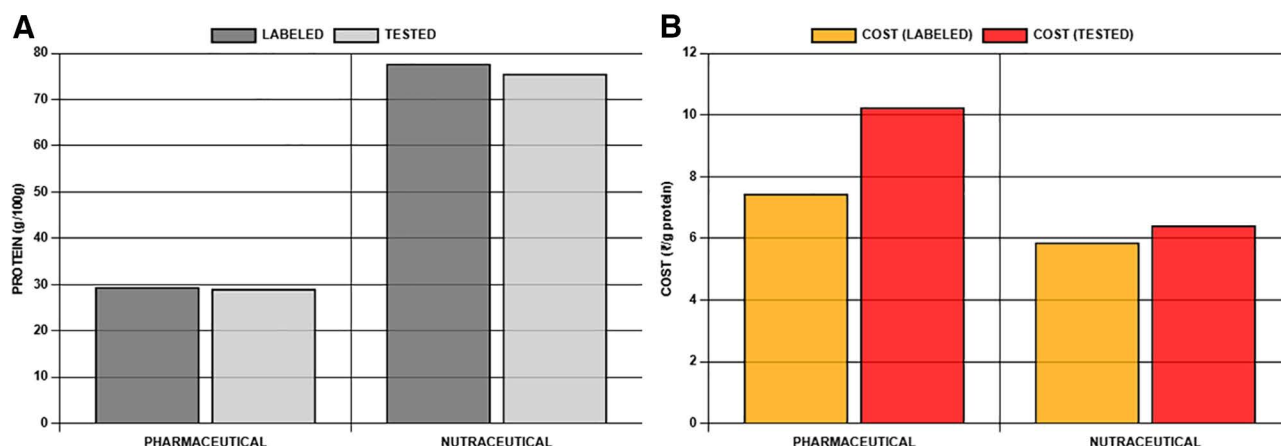


Figure 6. The analysis reveals superior cost-effectiveness in nutraceutical products. The cost effectiveness became even more pronounced when labeled (A) versus actual protein content (B) was considered, with several products showing cost increases exceeding 100% when calculated using tested versus labeled protein values.

cohort supplied a greater caloric load with limited high-quality protein, whereas the nutraceutical cohort delivered protein of substantially better quality and purity.

The unique, crowd-funded nature of this project warrants special mention as a model for future public health research. In a market where both regulatory oversight and industry self-reporting have proven insufficient, citizen science emerges as a powerful mechanism for independent verification and public accountability. This approach democratizes the scientific process, engaging the public as stakeholders in their own health and safety. By creating a transparent, open-access repository of

analytical data, our project provides a scalable framework that can be applied to other consumer product sectors where quality control is a concern. It demonstrates that when traditional oversight mechanisms falter, community-driven initiatives can effectively fill the void, empower consumers and provide regulators with the unbiased data needed to enforce standards.

Muscle and clinical recovery protocols typically aim for ≥ 20 – 25 g high-quality protein per serving. This is plausible when the powder contains ≥ 60 g protein/100 g. The nutraceutical cohort met this standard in 15/16 products, while 17/18 pharmaceutical brands failed. This mirrors wider surveys

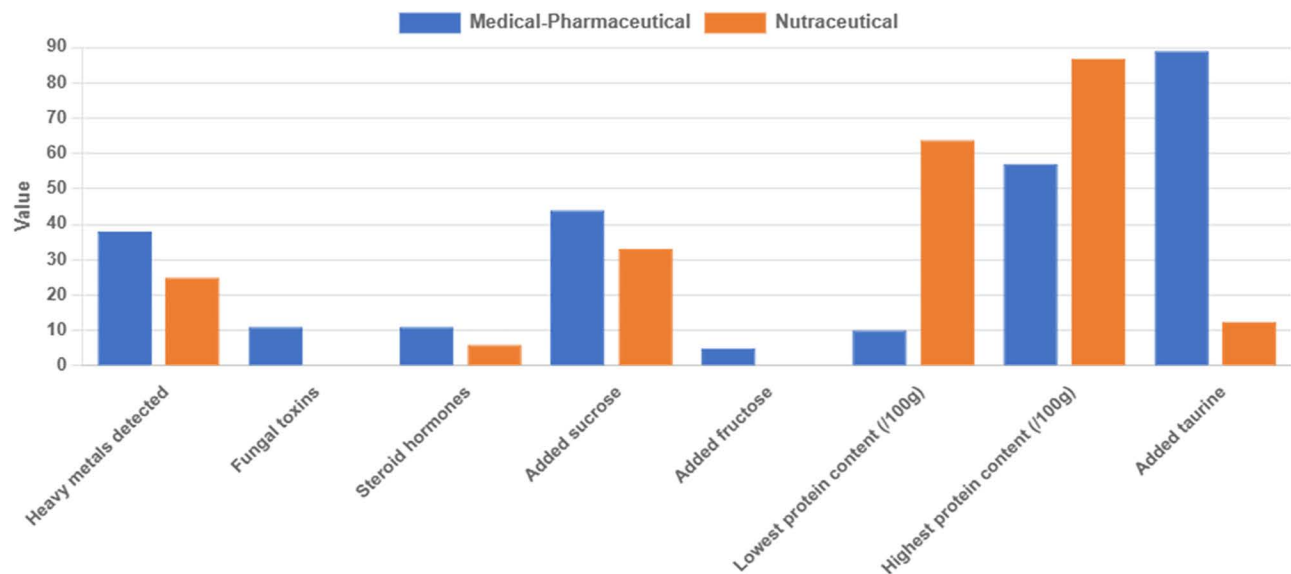


Figure 7. General outlook on the superiority of nutraceutical protein supplements when compared to medical pharmaceutical marketed protein supplements. The Y-axis demonstrates percentage except for lowest and highest protein content, which is amount in g per 100 g.

showing chronic label inaccuracy in sports foods where mean protein shortfall of 11% to 17% compared with claims have been described.^[11] For clinicians, that translates into under-dosing sarcopenic or post-operative patients unless extra scoops or alternative foods are prescribed. For consumers, it borders on deceptive marketing and fraud. Regulators already require $\pm 5\%$ to 10% accuracy in nutrition facts given the magnitude of errors observed, enforcement rather than new rules appears to be the missing link from our study conclusions.

Essential amino acids, which cannot be synthesized by the human body and must be obtained through diet, serve as a critical quality indicator for protein supplements. The analysis revealed that nutraceutical products not only contain higher absolute amounts of EAAs but also maintained better essential amino acid profiles. The top-performing products in terms of EAA content were predominantly from the nutraceutical category, with products achieving EAA contents ranging from 35.44 to 38.10 g/100g. In contrast, the highest-performing pharmaceutical product contained only 22.39 g/100g of EAAs. Six nutraceutical powders exceeded the “elite” target of ≥ 40 g EAA and ≥ 7 g leucine per 100g while none of the pharmaceutical products came close. The leucine “trigger” for maximizing muscle-protein-synthesis is around 2.5 to 3 g per dose, so a 30 g scoop of many medical-nutrition brands would not even clear this threshold. The quality of protein, as reflected by the amino acid profile, appears to be superior in nutraceutical products not just in quantity but also in balance. A well-balanced amino acid profile that closely matches human requirements is essential for optimal protein utilization. The data indicates that nutraceutical products generally provide more complete amino acid profiles, which would translate to higher protein quality scores and better nutritional value for consumers seeking protein supplementation. Clinicians and dietitians should therefore verify the leucine content on protein supplements prescribed for patients, not just grams of “total protein” when designing peri-operative or clinical care regimens.^[12]

Forward-looking practice will likely move toward leucine-standardized prescriptions rather than crude protein grams. A Spanish study showed that, in analyzed supplements protein content averaged 70.9% (range: 18–92.3%), with significant discrepancies between labeled and actual values and that monitoring and regulation of product quality and safety were necessary due to frequent labeling noncompliance.^[13]

Nearly half of pharmaceutical powders contained ≥ 2 g sucrose or fructose per 100g, sometimes unlabeled while several also relied on maltodextrin for bulk. Popular hospital-formulary protein drinks and powder supplements pack ≥ 14 g added sugar per pack or bottle. In hyper-catabolic or cachectic patients that carbohydrate may be justified, but for insulin-resistant, bariatric, or metabolic-disorders and liver-disease populations it may prove counter-productive.^[14] From a patient-centric view, a “high-protein” claim is meaningless if glycaemic load is high and brands should disclose added sugars prominently, and clinicians must scrutinize them.

The heavy-metal results from our study echoed recent broad surveys in which 47 % of market-leading powders exceeded levels for at least one toxic metal.^[3] A formal risk-assessment showed typical intakes may keep lifetime hazard below regulatory cut-offs, yet that comfort margin shrinks in special populations such as renally-impaired or pregnant patients.^[6] In another study, researchers examined 47 whey-based protein powders to see which minerals and metals they provide and whether typical or heavy use might be risky. A standard 30-gram scoop supplied helpful amounts of nutrients such as potassium and calcium and met about one-fifth of the daily needs for chromium and molybdenum while remaining safely below harmful levels for every metal measured. However, consuming 100g per day, especially the brands with the highest metal content, could push intakes above recommended limits. The authors recommend clearer labeling and stronger regulation so consumers and health professionals can better balance the nutritional benefits against the possible risks of heavy, prolonged whey-protein use.^[15]

The mean copper content in the pharmaceutical powders was roughly three-fold higher than in nutraceutical powders, and the single highest value exceeded the nutraceutical maximum by an order of magnitude. Nonetheless, the lower end of the pharmaceutical range overlapped completely with the nutraceutical results, indicating that elevated copper is not ubiquitous within either category. The presence of several below limit of quantification results in both groups shows that copper contamination is not inherent to whey protein production but likely reflected specific raw materials or processing aids. These findings provide quantitative evidence that, within this sample set, pharmaceutical-grade formulations tended to carry higher copper burdens than nutraceutical powders, warranting closer source-control scrutiny for products intended for clinical use.^[16]

Two pharma powders also harbored aflatoxin and progesterone, findings not isolated to this study: a previous Indian market data reported mycotoxins in ~14% of brands analyzed.^[1] Presence of progesterone in milk is acceptable, but it is usually not found in well processed whey, which triggers consideration for utilization of poor manufacturing and inadequate processing methods. For frontline doctors, unexplained transaminase flares or endocrine changes in supplement users warrant tox-screening and product tracing. Consumers should look for third-party badges and batch-specific contaminant certificates.

Taurine turned up in 89% of pharmaceutical powders but at milligram, not gram doses, far below levels used in cardiovascular or liver trials. Industry insiders concede the practice exists to inflate Kjeldahl nitrogen assays to “improve” protein content labeling.^[1,4] More than actual protein inflation, which was unlikely with the detectable doses, the absence of disclosure on taurine addition presents ethical mishaps for the manufacturer. From a practice point of view, a quick scan of the amino-acid panel for anomalously high glycine/taurine is the easiest red-flag for adulteration. Going forward, adopting high-performance amino-acid profiling as a regulatory requirement would close this loophole.

Based on the comprehensive cost-effectiveness analysis of protein supplements, significant distinctions emerged between pharmaceutical and nutraceutical product categories that warrant detailed examination. The data reveals fundamental differences in protein content, pricing strategies, quality control measures, and overall value propositions that have important implications for various stakeholders in the healthcare and nutrition sectors. Pharmaceutical companies may need to better communicate the value proposition of their products beyond protein content, emphasizing therapeutic benefits, clinical evidence, and specialized formulations that justify premium pricing. Alternatively, they might consider developing high-protein variants specifically designed to compete with nutraceutical products while maintaining their pharmaceutical-grade quality standards. Nutraceutical companies, while enjoying cost advantages, face the challenge of improving quality control and label accuracy to maintain consumer trust and potentially avoid future regulatory interventions.

From a stakeholder standpoint, these results suggest clear, practical steps. Consumers should favor whey isolates or concentrates that list a complete amino-acid profile, carry recognized third-party quality seals, and contain no more than about 1g added sugar per serving, rather than relying on vague “doctor-recommended” claims. Clinicians are advised to prescribe supplements based on verified leucine or total essential-amino-acid content rather than headline protein grams and should request certificates of analysis, particularly when treating renal, oncologic or hepatic patients, who are sensitive to both protein adequacy and contaminants. Nutritionists and dietitians should ask manufacturers for batch-specific heavy-metal and mycotoxin data and may choose whole-food fortifiers (e.g., milk powder, Greek yoghurt) when high-sugar powders threaten glycaemic control. Finally, regulators could focus less on writing new rules and more on enforcing current ones through regular post-market testing, mandatory disclosure of amino-acid profiles, and stronger penalties for labels that deviate by more than $\pm 5\%$ from laboratory values. The findings underscore the complexity of the protein supplement market and the need for consumers, healthcare providers, and regulators to consider multiple factors beyond simple protein content when evaluating these products.

Our findings also serve as a direct call to action for the manufacturers of medical-pharmaceutical protein supplements. To align their products with their intended therapeutic purpose, particularly for managing conditions like sarcopenia and post-operative recovery which are prevalent in the Indian population, several specific improvements are imperative. First,

formulations must be re-engineered to significantly increase protein density, aiming for higher per 100g standard met by most nutraceuticals. Second, protein quality must be prioritized by ensuring a minimum leucine content of 2.5 to 3g per serving to effectively trigger muscle protein synthesis, a critical factor for clinical efficacy. Third, the reliance on simple sugars and maltodextrin should be drastically reduced to avoid counterproductive glycaemic loads in vulnerable patient populations, such as those with metabolic disorders or liver disease. Finally, manufacturers must commit to pharmaceutical-grade quality control, including rigorous screening for contaminants and the cessation of undisclosed filler use, such as taurine. Adopting transparent, leucine-standardized labeling and providing batch-specific certificates of analysis would not only rectify the deficiencies identified in our study but also restore clinical confidence in products marketed for therapeutic use.

5. Limitations

We must also acknowledge the limitations of our study. Our analysis was based on a single batch for each of the 34 products. While this provides a critical snapshot of the market at the time of purchase, we recognize that product composition may vary between batches due to fluctuations in raw material quality and manufacturing processes. Therefore, our findings for any single product may not be representative of all batches over time. However, the stark and consistent categorical differences observed between the pharmaceutical and nutraceutical groups suggest that the quality gap is a systemic issue rather than an anomaly of batch selection. Future regulatory and independent surveillance efforts should incorporate longitudinal, multi-batch testing to assess the consistency of product quality and hold manufacturers accountable for maintaining standards over the long term.

6. Conclusion

Our study results call for regulators to ensure that products marketed for clinical nutrition meet, at minimum, the compositional standards achieved by sports-nutrition supplements. Routine post-market surveillance with full amino-acid and contaminant panels, public certificates of analysis and mandatory disclosure of added sugars and non-protein nitrogen sources would close the current quality gap. Leucine-standardized labeling could help clinicians tailor protein prescriptions for sarcopenia, peri-operative recovery and chronic liver disease, while blockchain-based supply chains would improve traceability. Future work should extend similar scrutiny to plant and blended proteins, evaluate bioavailability in vulnerable patients and assess long-term outcomes of chronically under-dosed “medical” powders. Public-private funding models, such as the one used here, offer a scalable framework for independent quality audits and open-access databases that empower consumers, inform clinical guidelines and guide policy. Transparent analytical data enforced labeling accuracy, and accessible education will be crucial for protein supplements to fulfill their promised therapeutic and nutritional roles.

Until faster on-site verification and transparent supply chains are commonplace, an independent analysis driven by “citizen science” and public participation remains the most dependable marker of product quality in current landscape of nutritional supplements.

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